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How autonomous research shapes scientific understanding

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Autonomous science promises to translate experimental agency into advances in scientific understanding. Yet how agents organize inquiry and turn experimental experience into knowledge remains poorly understood. Here we introduce ChemWorld, programmable chemical worlds combining autonomous experimentation with independent control of underlying laws and supplied information. Across chemical systems, model configurations and repeated investigations, we connect research objectives and prior knowledge to experimental choices, evolving explanations and subsequent judgments. Agents propose and test explanatory possibilities, establish useful local relationships and retract unsupported expectations. Similar predictive accuracy can nevertheless rest on different evidence histories, while successful local research can leave mechanisms and wider applications unresolved. Matched chemical worlds distinguish the information acquired by an experiment from the interpretation a researcher gives it; research-opportunity comparisons examine the consequences of conducting inquiry. Together, these studies make autonomous research itself an object of empirical investigation, linking how exploration is organized to how scientific understanding forms, develops and guides further inquiry.

Scientific discovery involves more than deriving answers from established premises: it also introduces ways of understanding phenomena that open new directions for inquiry. Einstein began his 1905 paper on special relativity with a tension between the observable effects of magnet–conductor relative motion and the asymmetry of their prevailing theoretical descriptions. He introduced principles from which consequences for simultaneity, time and motion could be derived[1]. This episode illustrates the creative work of reconsidering the premises through which the world is understood.

Abduction concerns a central part of this activity: proposing explanatory hypotheses worth investigating. In the Peircean account, abduction introduces possible explanations, deduction develops their testable consequences, and empirical testing and induction assess what should be retained, revised or abandoned[2]. An explanation can thus be both a product of research and a starting point for further exploration. Experiments evaluate existing judgments while also changing which phenomena demand explanation and which questions become worth pursuing.

An autonomous researcher does more than act on the world. By choosing experiments and measurements, it also shapes the experience from which its scientific judgments form: which phenomena become visible, which explanations encounter a test and which results guide the next investigation. As AI scientists and self-driving laboratories connect increasingly elaborate experimental cycles, this organization of inquiry becomes part of the scientific activity delegated to machines[3,4]. The promise of autonomous science therefore concerns both the expansion of experimental capability and the transformation of experimental experience into scientific understanding.

Robot scientists and self-driving laboratories have demonstrated hypothesis testing, materials discovery and reaction optimization[5–8]. Language-model agents extend the connection between scientific instructions, tools and laboratory operations[9,10]. These advances make it possible to investigate a further question: how does an agent's own research change the relationships it uses to explain and predict the world? A successful procedure establishes an experimental achievement, and a correct forecast establishes predictive accuracy. Neither alone reveals the evidence from which a judgment arose or the scope that evidence supports[11,12].

The difficulty is that the evidence available to an autonomous researcher is itself an outcome of its research. An initial explanation can suggest a productive experiment; the observation can support, narrow or displace that explanation; the resulting judgment can then guide another intervention. A sequence of successful experiments may establish a useful local relationship without resolving how it extends beyond the observed conditions. Conversely, an unsuccessful intervention can advance understanding by ruling out an expectation. Studying this interaction requires connecting what the researcher initially had reason to expect, what its experiments actually tested and what it subsequently treated as established[13–15].

We study these connections in ChemWorld. Our central question is how AI organizes inquiry to develop scientific understanding when its initial information does not determine the answer, and how developing understanding in turn shapes exploration. Four complementary experimental matrices examine research objectives and information, the formation of judgments, access to research, and discrimination between matched chemical worlds. Together they cover six chemical systems and connect repeated outcomes to complete public research histories. Historical model and budget comparisons extend this perspective to different routes of exploration. The resulting evidence distinguishes what an experiment accomplishes, what relationship it supports and how that relationship enters further inquiry.

Making autonomous scientific inquiry experimentally tractable

ChemWorld combines materials, operations and instruments into stateful virtual chemical systems governed by thermodynamic, kinetic and coupled process relationships. Reaction, thermal processing, phase equilibrium, separation and crystallization components can be connected into multistage procedures. Sample states persist, so an upstream reaction or thermal history can change a later separation or measurement. Agents choose quantities, processing sequences, measurements, repetitions and subsequent experiments within fixed resource allowances. Their choices determine both the procedures they develop and the observations from which they reason. An evaluator independently controls world parameters and public information, records committed operations and obtains withheld reference outcomes (Fig. 1). The agent does not receive those outcomes during research or assessment.

This arrangement makes the research process experimentally accessible. Knowledge tests and supplied-data tasks assess important scientific abilities[16,17], and interactive discovery environments already connect action to observation[18–21]. ChemWorld connects evidence selection to evolving public judgments and their later use in composable chemical processes. An attempted operation can be rejected without changing the sample; an observation can support a relationship locally without establishing it under every compatible intervention. Execution records and independently evaluated world responses allow these distinctions to be checked.

The integrated design contains 450 planned investigations across four matrices. The objective–information matrix comprises ninety electrochemical and ninety reaction-processing sessions: five worlds, three information conditions, two research commissions and three fresh sessions per cell. The process matrix comprises forty-five sessions each in partitioning, purification, aqueous equilibrium and reaction-coupled crystallization. The opportunity matrix compares research with no research in equilibrium and crystallization, thirty sessions per system. Finally, thirty investigations in matched aqueous worlds connect the physical information acquired by an experiment to its interpretation. The latest block supplies 300 sessions; 150 are retained from earlier fixed protocols. This is a common coverage design with distinct experimental blocks, not a pooled trial conducted under identical protocols (Extended Data Fig. 2).

The information conditions alter what researchers initially know. Opaque retains the public task and operating contract without an additional instance dossier. Aligned supplies correct but incomplete information, and MisIndexed supplies a specified mismatch. Discovery commissions emphasize investigating scientific relationships; optimization commissions emphasize developing a successful procedure. Within each fixed block, model configuration, interface and experimental allowance are held to the corresponding design. Fresh sessions share fixed worlds and observation keys; the model sampling seed is not controlled. These contrasts examine different starting conditions for inquiry without treating worlds or repetitions as independent samples of science in general.

All 300 new sessions reach a terminal state: 288 complete the full chain, 298 provide valid forecasts, and 3,379 of 3,420 planned source assays are obtained. Ten purification sessions exhaust material and miss forty-one assays; two reaction sessions have an absent or invalid forecast. Valid predictions from incomplete source histories remain available, with their shortfalls identified. Four service interruptions are resolved by technical retries, retaining the interrupted attempts. Scientific failures are not replaced. The new records contain 873 notebook versions and 31,922 public operations, all read alongside the available sealed accounts and predictions; 1,363 quotation anchors are verified. These counts establish coverage, not rates of discovery or abduction.

Notes locate judgments before and after observations. After research, agents seal a scientific account, predict twelve new interventions without laboratory access and reflect without reference truth. We trace the phenomenon, proposed explanation or expectation, intended test, actual operations, subsequent revision and later use. Terminal revisions and unexecuted plans remain part of the record. Three quantities remain distinct throughout: what is true in the programmed world, what the information available at a given time supports, and what the agent subsequently states or uses. Historical campaigns, including a five-model equilibrium comparison and separate budget studies, supply additional contrasts without enlarging the current matrix denominators.

The matched-world matrix tests one further distinction. Five pairs of aqueous worlds share public framing and selected parameters but differ in whether precipitation is coupled to an additional neutral aqueous ion pair. Two supplied ultradilute assays have closely matched underlying responses; autonomous loading, dilution and scale choices can reveal larger differences. Thirty complete investigations yield 360 final assays and eighty notes. Replaying each chosen action sequence in its paired world shows that every session acquires at least one observation differing by three assay standard deviations in two response channels. This establishes available physical information, not unique identification of the mechanism or thirty successful abductions.

The self-organization and adaptation of scientific inquiry

Research begins with goals and information, but its scientific content develops through the choices they elicit. The repeated electrochemical matrix makes this distinction visible (Fig. 2a,b). Averaging sessions within worlds and then weighting worlds equally, optimization rather than discovery increases the retested recommendation score from 0.4813 to 0.5852 and selective-product yield from 0.1310 to 0.4225. Yet prediction errors increase for score (0.1539 to 0.1829), yield (0.0665 to 0.0781) and Faradaic efficiency (0.1677 to 0.2218), each in all five worlds. These results establish a separation between developing an effective proce-

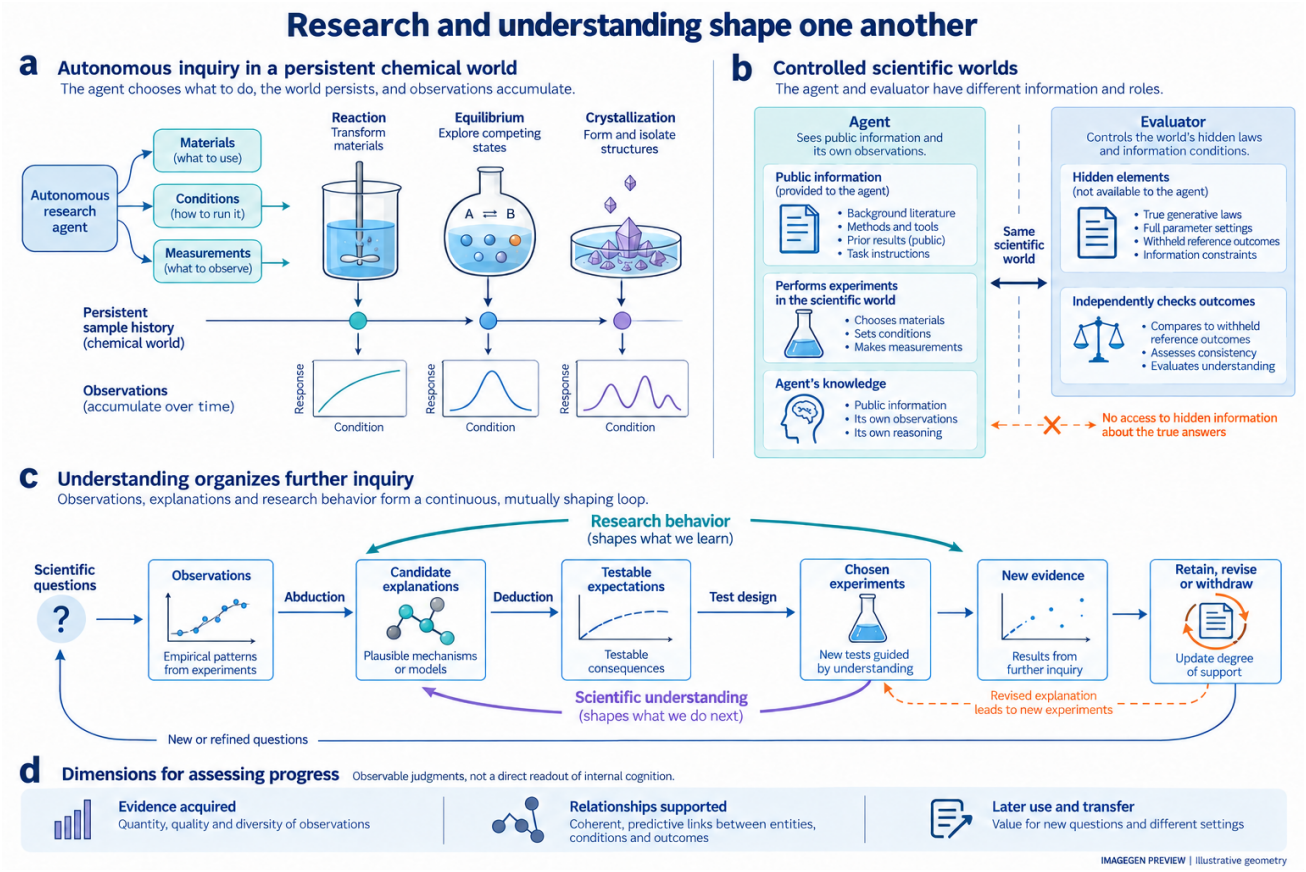


Figure 1 | Research and understanding shape one another. a, Choices and persistent sample histories determine available observations. b, Agent-visible information is separated from evaluator control and withheld references. c, Abduction proposes explanatory possibilities; deduction develops testable consequences; experiments and revision can organize further inquiry. d, Evidence acquisition, supported relationships and later use are distinct dimensions of progress. The cycle is an analytical framework for public records, not a direct measurement of cognition or a claim about every action. ImageGen preview: geometry is illustrative; exact values are in the retained evidence.

dure and judging new interventions. They do not establish that operating success itself causes misunderstanding.

The process histories show how both achievements and limits arise. One optimization researcher finds that negative potential outperforms positive potential at the tested magnitude, then improves materials and duration on the negative side. Its terminal note explicitly declines to call the result a universal sign law. Nevertheless, subsequent forecasts assign near-background responses to untested positive potentials inside the investigated magnitude, and their intervals miss the corresponding references. Another optimization session prospectively challenges absolute polarity specificity, executes a matched reverse-polarity test with yield 0.384, and revises the account to polarity advantage rather than exclusivity. Its forecasts retain reverse-polarity production. Optimization can therefore include genuine explanatory tests; the aggregate contrast is not a classification of all optimization as self-confirmation.

Information changes the consequences of inquiry in a similarly differentiated way (Fig. 2c,d). Under the reaction optimization commission, Aligned rather than Opaque information improves recommendation score from 0.2735 to 0.4429 and yield from 0.5502 to 0.7116. Risk prediction error falls from 0.2389 to 0.1226, while conversion error rises from 0.0355 to 0.1595. All four directions occur in all five worlds. The comparison concerns the complete information package; it does not isolate a particular sentence or identify a single intervening belief. A single error average across reaction responses would obscure this divergence.

The organization of research also depends on what an intermediate result makes worth doing next. In purification, a researcher has a no-wash procedure with purity 0.814 and recovery 0.0479, and a higher-phase-amount procedure with better recovery but purity 0.7886. Before the next batch it predicts that a 0.005 L wash can bring the latter above 0.80, specifying that it will repeat this alternative only if recovery also exceeds 0.0479. The wash produces purity 0.8013 and recovery 0.0441. The researcher then returns to the original no-wash procedure for a repeat, obtaining 0.8149 and 0.0368. The experiment partly meets its scientific expectation but fails its prospective action criterion; the next choice follows that distinction. The repeat also exposes variability rather than establishing a global optimum.

An explanation can change the evidence subsequently available. After low recovery, another purification researcher suspects a missing feed component and adds material through two operations that consume the same shared stock. The altered procedure exhausts that stock, preventing later planned temperature tests. Its final account correctly recognizes that those unperformed tests do not

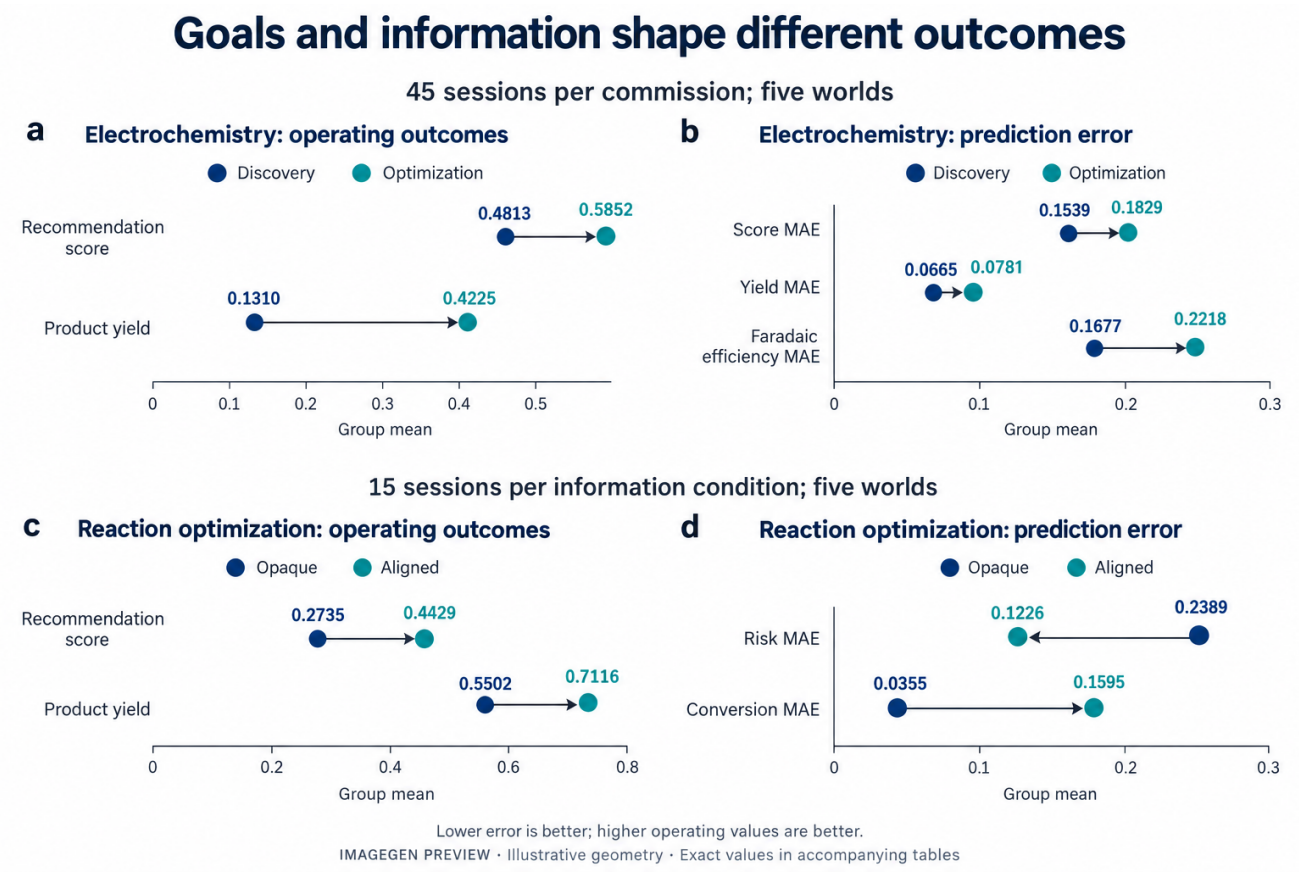


Figure 2 | Goals and information have different consequences for operation and prediction. a,b, Repeated electrochemical discovery–optimization comparisons, forty-five sessions per commission across five worlds and three information conditions. Retested scores and yields improve under optimization while three prediction errors increase. c,d, Reaction optimization under Opaque and Aligned information, fifteen sessions per arm: recommendation score and yield improve, risk error decreases and conversion error increases. Means average sessions within worlds, then worlds equally. Arrows connect group means, not the development of one agent. ImageGen preview: geometry is illustrative; numerical labels refer to retained evidence.

support a temperature relationship. Here a proposed explanation enters an action, the action changes resources, and resources change the remaining opportunity to observe. This is an observable route by which understanding shapes exploration, even though the interpretation concerns both chemistry and operation semantics.

Different histories can also lead to similarly accurate judgments. In the historical five-model equilibrium comparison, eleven of Astra's fifteen campaigns assay positively loaded samples at nominal concentrations no greater than 1 mM, compared with none of fifteen for each of Sol, GPT-5.5, Luna and Terra (Fig. 3). One Astra campaign first observes dissociation of 6.3–7.4%, then enters an 18.5 micromolar condition and observes 53.8%. Its account distinguishes the local plateau from the dilute response, and predicts 58.80% at 13.3 micromolar, with an 80% interval of 54.20–63.40%, against a reference mean of 57.97%. Sol/Opaque instead achieves dilute macro error 0.01833 without directly sampling that region. Acquiring a new phenomenon and making a successful extrapolation are distinct achievements; direct sampling of every target is not required for useful inference. These historical observations describe particular campaigns, not fixed model traits.

Matched chemical worlds reveal how an explanatory possibility can organize more specific evidence acquisition (Fig. 4). One researcher proposes that a sharp precipitation onset with a buffered dissolved response would support a solubility-limited interpretation, whereas smooth changes would also admit an empirical partition description. After six assays, it proposes denser concentration sampling, a central repeat and increasing loading. The next six batches implement these investigations. The final account retains a smooth, monotone relationship but leaves a broadened threshold and an empirical partition description unresolved. Its forecast states: "Precipitation estimates use a smooth monotone concentration response because no sharp threshold was resolved."

A second session tests whether diluting a concentrated sample reverses the high-loading response. It reduces nominal loading from 0.75 to 0.375 M: normalized pH rises from 0.08045 to 0.10803, and the final precipitation signal is 0.08026 compared with a pre-dilution measurement of 0.10004. The researcher supports approximate reversibility while retaining uncertainty about small hysteresis. The nearest separately prepared comparator is 0.4 M, and the precipitation readings use different instruments; this is not an exact hysteresis determination. The scientific advance lies in a stated alternative becoming a prospective test and then a bounded judgment.

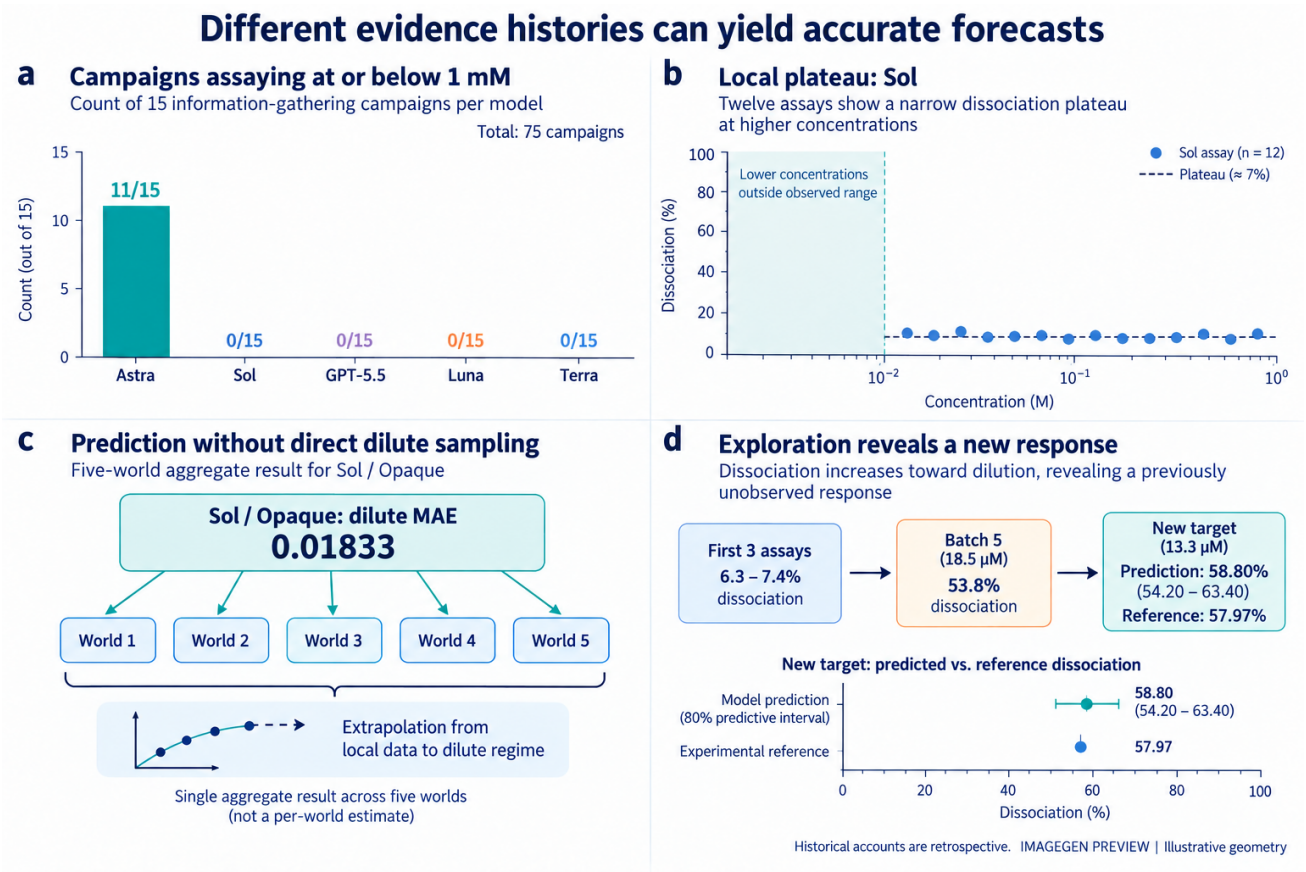


Figure 3 | Different routes from evidence to prediction. a, Campaigns reaching at or below 1 mM among fifteen per model. b, An illustrative local dissociation plateau. c, The five-world Sol/Opaque dilute MAE is an aggregate, not a separate value for each world. d, Selected stages of one Astra campaign and its sealed forecast, with the original 80% predictive interval. Historical scientific accounts are retrospective. These observations do not establish fixed model traits. ImageGen preview: geometry is illustrative; exact values are in the retained evidence.

The formation and evolution of scientific understanding through experimentation

The scientific content of an investigation becomes visible when a proposed relationship encounters a test and changes what the researcher is prepared to claim or do. In reaction processing, a researcher observes continued conversion during a post-heating wait, despite cooling by approximately 44 K. Before the next comparison it predicts that quenching before the same wait will leave conversion near 0.42 rather than the approximately 0.62 observed without immediate quenching. The executed test produces 0.427. This connects an observed discrepancy, a testable explanation and a successful local prediction (Fig. 5a). It does not establish all subsequent solvent or temperature extrapolations.

Another reaction investigation distinguishes chemical progress from the risk reported at the end of processing. An anticipated quench contrast changes final risk while leaving the chemical effect unresolved; the researcher narrows its claim accordingly and uses the final-state relationship in later predictions. A separate time series at 410 K observes increasing conversion alongside declining selectivity, with product yield peaking at the intermediate duration. The agent retains the local trade-off while leaving the reaction-network topology open. Useful scientific understanding can therefore take the form of a constrained relationship before a unique mechanistic account is available. Quench changes temperature, risk and the clock but does not integrate additional chemical reaction in this implementation; waiting before quench does. Measurements from different instruments cannot by themselves establish the intervening chemistry.

Prospective tests also narrow explanations when they fail. In partitioning, a researcher explicitly treats a near-plateau under ordinary mixing as provisional. Weak mixing gives an organic fraction of 0.536, ordinary mixing gives 0.631 and 0.662, and a stronger condition gives 0.732. The stronger condition defeats the anticipated plateau, which the subsequent account withdraws (Fig. 5b). Because duration and rotation rate change together, the evidence supports a revised mixing relationship rather than separately identifying their effects or a unique transport mechanism.

Crystallization offers both successful and unsuccessful continuations (Fig. 5c,d). One researcher prospectively adds a third thermal cycle, obtaining fines 0.494 and recovery 0.584; a fourth cycle does not continue the improvement, so it narrows the monotonic expectation. A measurement before filtration gives fines 0.506, emphasizing that the subsequent value just below 0.50 is not robust threshold evidence. Another session lowers reagent charge and reduces fines to 0.583, then predicts that reducing it further to 0.005 mol will produce an acceptable result. Instead the next batch

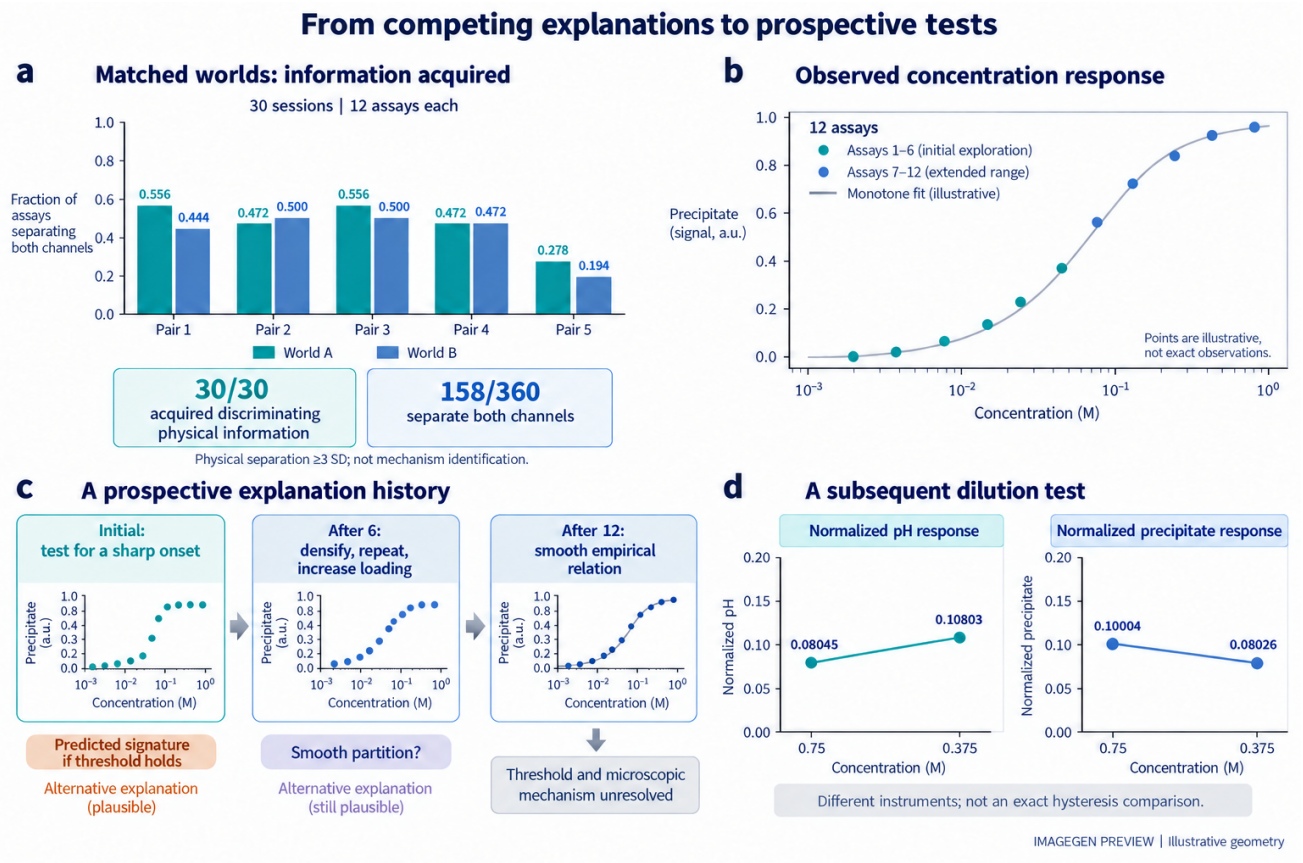


Figure 4 | Explanations motivate tests and bounded relationships. a, Mean fractions of twelve assays separating paired worlds in at least two channels by three assay SD, averaged across three sessions per world; physical information is not mechanism identification. b,c, Illustrative concentration responses and the recorded progression from conditional expectations to denser sampling, repetition and a bounded empirical relation. d, Another session's prospective dilution test, with retained response values. The readings use different instruments and do not constitute an exact hysteresis comparison. ImageGen preview: geometry is illustrative; exact values are in the retained evidence.

lacks recoverable particles, filtration rolls back and final recovery is only 0.0000445. The account narrows the useful region to a local trade-off between fines and recovery and acknowledges that no batch met the target. Scientific revision here does not depend on an eventual operating success.

These histories complement earlier corrected-crystallization studies in which lowering charge establishes a local improvement and a solvent-volume interpolation is explicitly retracted after failure. They also show why the relevant unit is a judgment rather than an entire session. The session that correctly retracts its dilution prediction elsewhere describes an unsuccessful 1 g seed test, although only 0.05 g of seed had committed. A defensible revision and an unsupported empirical premise can coexist in the same investigation. Scientific vocabulary, task success and the number of hypotheses proposed do not distinguish them.

The reconstruction of experience is itself consequential (Extended Data Fig. 6). In one reaction session, two 600 s heating attempts roll back with no elapsed reaction time or temperature change. The terminal account nevertheless treats them as 1,200 s without reaction, proposing an induction or history effect and carrying that premise into later forecasts. Another session explicitly excludes rolled-back heating from its chemical interpretation. In crystallization, one intermediate note similarly turns 0.05 g of committed seed into a claimed 1 g test and uses it to plan a 1.5 g test. High-dose requests continue to fail, while the account later describes larger tested amounts. Across the fifteen new research sessions, eight contain seed-dose misstatements associated with rejected additions, but their timing and subsequent uses differ. This is a specific execution–account discrepancy, not a single established psychological mechanism.

These errors require interpretation against the implemented experiment. Some returned envelopes describe an experiment as completed while an inner physical transition is rolled back, and distinct operations expose different legal limits. In one reaction session, the researcher applies the controller's 430 K limit to direct heating, which is publicly allowed up to 470 K, then predicts near-zero outcomes for requests it expects to roll back. Such cases concern the relationship between operation semantics, remembered evidence and scientific judgment; they cannot all be attributed to misunderstanding chemical kinetics. Conversely, a terminal false account cannot be assumed to have directed earlier source actions. We distinguish intermediate premises with subsequent actions from explanations first recorded after research.

Local support and later application can also diverge without a false execution history. A reaction researcher repeats

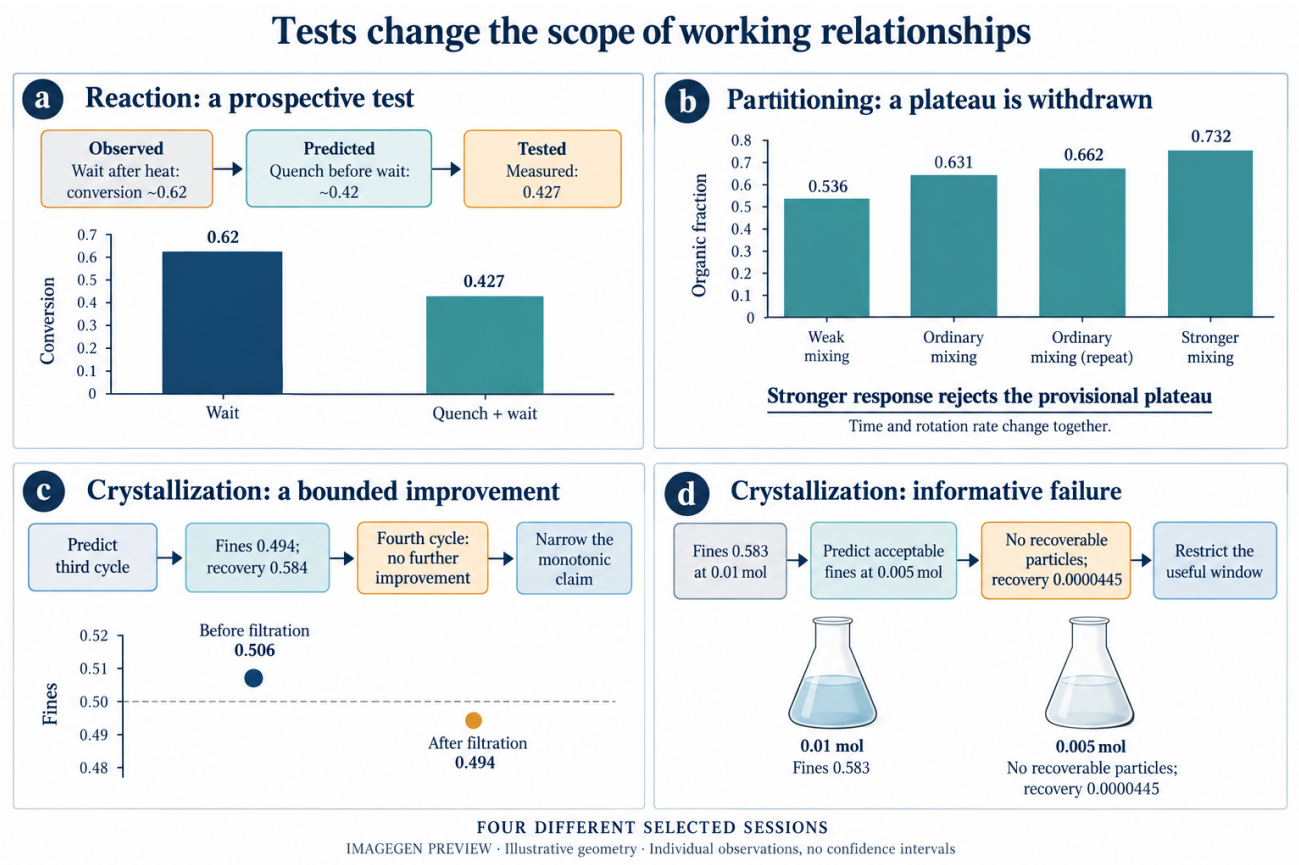


Figure 5 | Tests change the scope of working relationships. a, A prospective quench-before-wait prediction is followed by conversion 0.427, near the predicted 0.42. b, A stronger mixing condition leads to withdrawal of a provisional partitioning plateau; duration and rotation rate are confounded. c, A third crystallization cycle gives a near-threshold result, while a fourth fails to continue the improvement. d, Further dilution fails to preserve recoverable particles and narrows the proposed operating window. The panels are separate selected histories. Values are individual observations, not estimated treatment means or confidence intervals. ImageGen preview: geometry is illustrative; exact records accompany the manuscript.

a 360 K, 3,600 s procedure, obtaining scores 0.34983 and 0.34779, and observes that longer heating increases conversion while decreasing product yield. This supports a local time trade-off. Its later forecasts extend severe product loss to untested conditions: conversion error is 0.00345, but yield error is 0.31812. Accurate conversion does not validate the whole product-loss account. In equilibrium study A, a real dissociation plateau likewise enters an extreme-dilution forecast even though the agent acknowledges that the regime is untested (Extended Data Fig. 5). Its intended feed-identity contrast was never realized, because both commands added the same species. The empirical plateau and the proposed explanation therefore have different evidential standing.

Together, these cases reveal several transitions through which research becomes understanding: observations motivate a candidate; a candidate selects a test; a result supports or narrows a relationship; and that relationship enters another judgment. A transition can succeed for one claim and fail for another. Failed tests can produce warranted revisions, unresolved mechanisms can coexist with useful local knowledge, and a correct local finding can acquire an unsupported wider application. Tracing these distinctions makes the scientific content of inquiry visible beyond the correctness of its final report.

The nature and conditions of scientific progress in autonomous inquiry

The preceding histories establish that autonomous research contains substantive scientific activity. The opportunity matrix asks what difference access to that research makes under common initial information and assessment. Both systems compare twelve-batch research with no source experimentation in five fixed worlds, with three fresh sessions per arm and world. All sessions use Opaque information. The comparison estimates the complete opportunity to organize experiments, receive feedback and record judgments; it does not isolate abduction, autonomy relative to supplied data, or notebook use.

The new crystallization comparison provides clear positive evidence (Fig. 6a,b). All thirty sessions provide valid forecasts, and all 180 research assays complete. Relative to no research, world-equal mean absolute error falls from 0.13853 to 0.01661 for purity and from 0.23619 to 0.07641 for size index. Recovery error declines from 0.13532 to 0.11579 and fines error from 0.37683 to 0.30110. For purity, nominal 80% interval coverage rises from 0.46667 to 0.98333 while mean width contracts from 0.44744 to 0.09743. Real experimental experience can therefore establish and transfer a useful stable relationship;

weak retention of evidence is not a general account of crystallization research.

The same comparison also distinguishes point accuracy from uncertainty quality. For fines, interval coverage falls from 0.42778 to 0.35556 as width narrows from 0.68500 to 0.38450. The interval score, which penalizes both width and misses, worsens from 1.4794 to 1.8047, with this direction in all five worlds. Improved average point prediction does not ensure that uncertainty adequately represents unresolved conditions. The appropriate conclusion is response-specific: the research strongly improves purity judgments while its fines predictions retain substantial uncertainty failures.

The retained equilibrium comparison identifies a different boundary (Fig. 6c,d). Among fourteen valid research forecasts and fifteen no-research forecasts, overall normalized macro error falls from 0.132086 to 0.035632, with improvement in all five worlds. Yet error on the three prospectively designated extreme-dilution queries rises from 0.061467 to 0.103677, worsening in four worlds; on the other nine queries it falls from 0.155625 to 0.012950. Dissociation accounts for 85.2% of the dilute macro-error increase. Allowing the single missing research forecast any error in its legal range does not reverse either aggregate direction (Methods). The regional reversal is specific to these worlds and judgments, not an inevitable consequence of acquiring evidence.

Prior information likewise has no universal ranking. The partitioning matrix gives phase-fraction errors of 0.04654, 0.05621 and 0.07700 under Aligned, Opaque and MisIndexed information, respectively; Aligned improves on MisIndexed in all five worlds. The two fractions are complementary responses, not independent replications. Purification gives similar purity errors across the three conditions, approximately 0.152–0.154, while MisIndexed has the lowest recovery error. Material exhaustion also changes the amount of evidence acquired in ten sessions. Combined with the reaction contrast, these results show why a single aggregate information-benefit score would conceal scientifically different judgments.

Historical evidence provides additional conditions on these effects (Extended Data Fig. 1). In the original Sol equilibrium campaigns, correct partial information lowers error on nine queries from 0.01008 to 0.00465 but increases it on the three most dilute queries from 0.01833 to 0.15862. A retained repeated cohort shows the same aggregate directions, with different magnitudes: 0.00907 to 0.00512 and 0.09880 to 0.13205. Other model configurations do not all share this reversal. Historical twelve- and twenty-four-batch comparisons also improve some predictions without a corresponding improvement in feasible crystallization recommendations. These separate blocks identify dependencies on researcher, condition and resource envelope; they are not additional replications of the new matrix.

A complementary source-only diagnostic asks how fully acquired evidence constrains subsequent judgments. In forty-two valid sessions from the corrected process cohort, purity error is 0.03853 for the agent versus 0.00510 for the mean of its own source observations, whereas recovery error is 0.10129 versus 0.20443, with the agent better in thirty-nine sessions (Extended Data Fig. 3). Those source-only calculations diagnose information left unused in a given history. They are not no-research controls and do not show that experimentation lacks value. The new opportunity comparison establishes substantial purity gains relative to no research, while the retained diagnostic shows that a researcher can still underuse its available evidence. Both can be true without pooling the cohorts or assigning them the same baseline.

Finally, the matched-world investigations achieve errors of 0.00196 for normalized pH, 0.00312 for free dissociation fraction and 0.00243 for precipitation while some explanatory alternatives remain unresolved. Empirically established relationships can guide accurate later inference without unique mechanism identification. That block has no no-research arm and uses different chemical laws, so its low errors are not a causal improvement estimate against the earlier worlds. Across the four matrices, progress is therefore multidimensional: effective procedures, bounded relationships, successful transfer and well-supported uncertainty can develop at different rates within the same autonomous inquiry.

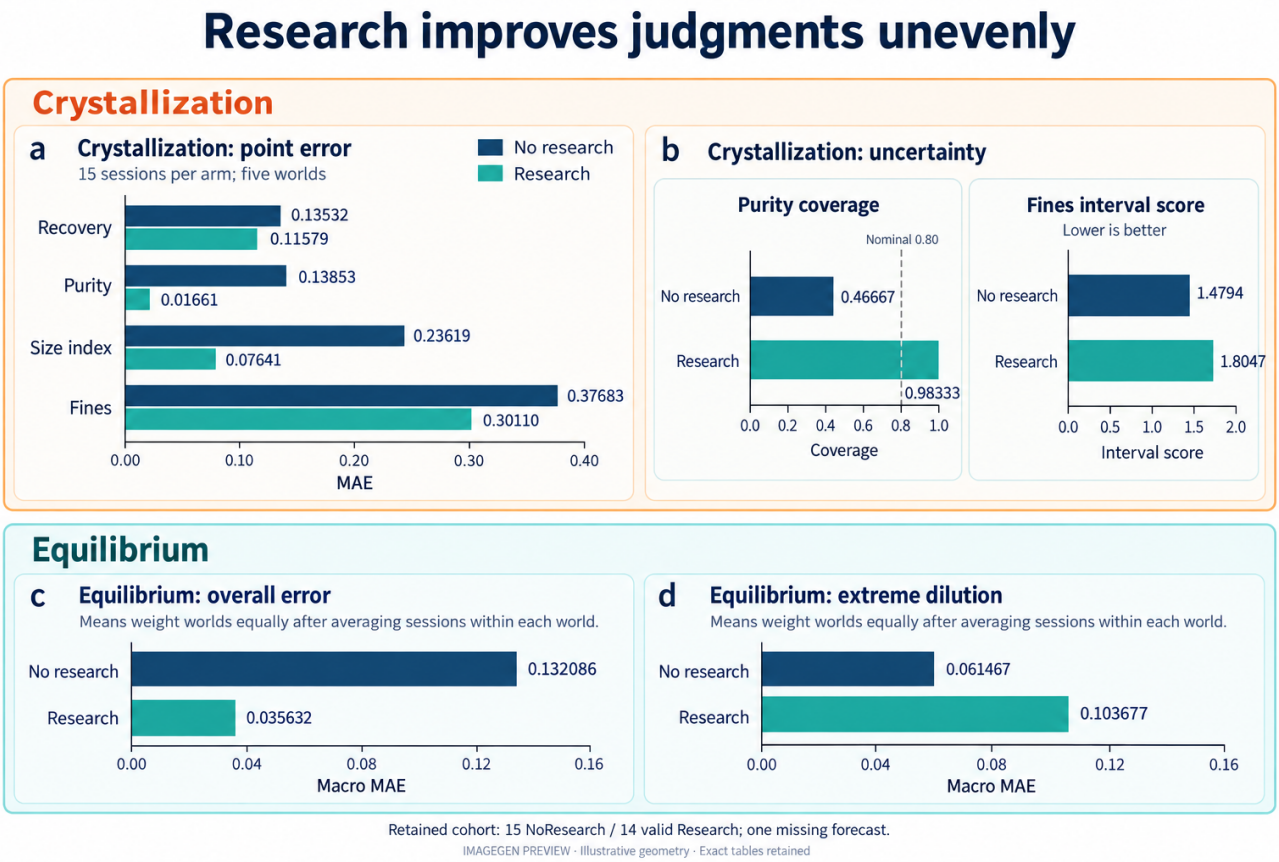


Figure 6 | Research changes judgments differently across responses and conditions. a, Crystallization point errors with and without research, fifteen valid sessions per arm. b, Purity coverage improves while fines interval scores worsen; coverage concerns nominal 80% prediction intervals, not confidence intervals on the group means. c,d, The retained equilibrium comparison improves overall prediction but worsens extreme-dilution prediction, with fourteen valid Research and fifteen NoResearch forecasts. Panels report separate experimental blocks, each averaging sessions within worlds and then five worlds equally. ImageGen preview: geometry is illustrative; exact tables and missingness are retained.

Discussion

Autonomous research changes both the evidence a machine encounters and the relationships it uses to understand that evidence. The four matrices connect this reciprocal process at complementary levels. Goals and supplied information alter the conditions under which inquiry unfolds; full histories show candidate explanations becoming tests and revisions; opportunity comparisons measure changes in subsequent judgment; and matched-world replay distinguishes available physical information from its interpretation. The contribution is an empirical account of how inquiry produces, qualifies and applies scientific relationships, rather than an assessment of whether an agent can complete an experimental cycle.

Abduction makes one creative aspect of this process explicit. A discrepancy can suggest an explanation that changes the next experiment, as in tests of post-heating reaction, precipitation responses and reversibility. Deductive consequences and empirical comparisons then change its standing. Other valuable activities include establishing a local relation, repeating a borderline result, retracting an interpolation and recognizing an unperformed test. These activities need not all be called abduction. Their integration shows how developing understanding participates in the actual organization of inquiry, including when a test fails or a unique mechanism remains unresolved.

The results distinguish truth, evidential support and subsequent use. An accurate forecast may rest on an unresolved explanation; a well-supported local relationship may be extended beyond its tested range; and a researcher may improve substantially while still failing to exploit all of its observations. The crystallization opportunity comparison is especially instructive: research enables strong purity and size predictions, while fines uncertainty remains unreliable. The older source-mean diagnostic asks a different question—how fully the available evidence is used. These differences are lost if each investigation receives one label of scientific success or failure[12,22,23].

The process analysis further locates a distinction between acquiring evidence and reconstructing what was tested. Rejected requests can enter later accounts as completed experiments, while other sessions explicitly exclude them. A public scope limitation can coexist with numerical forecasts that do not respect it. At the same time, agents can use counterexamples to abandon absolute claims, revise action criteria and delimit useful operating regions. These are judgment-specific developments within research histories, not mutually exclusive types of researcher. Interface semantics and the timing of public statements determine

how strongly each development can be attributed to the preceding research.

The controlled comparisons bound the explanations they support. Research-opportunity contrasts estimate the full research package, not an isolated effect of abduction or experimental autonomy. Matched-world replay holds actions fixed and does not estimate how different feedback would change the next action; physically discriminating observations do not prove unique mechanism identification. Goal and information comparisons do not identify a selected history as the sole mediator of an aggregate difference. A separate fixed-recipe quench intervention identifies physical effects without allowing agents to adapt (Extended Data Fig. 4). None of these designs requires a common self-confirmation mechanism or a successful correction strategy to explain why knowledge formation deserves empirical study.

The evidence is limited to programmable systems, fixed worlds and, for the repeated matrices, one model configuration. Fresh sessions are not controlled model-seed replications or independent samples of world laws. Public notes are not direct access to private cognition; missing intermediate notes limit temporal claims. All new histories are read, but illustrative judgments are selected retrospectively and interpretation has no independent human second-rater estimate. The integrated coverage retains different protocol periods and all scientific failures, including material exhaustion and invalid forecasts. The virtual chemistry and operation interfaces also require scrutiny when distinguishing a chemical interpretation from an execution error. These limits constrain generality without erasing the observed tests, revisions and response-specific effects.

Einstein's opening problem illustrates that scientific discovery can require reconsidering the premises of understanding; it is not a historical equivalence test for present agents. ChemWorld makes smaller, concrete instances of this broader activity observable. It connects experimental choices, acquired evidence, developing explanations and subsequent decisions within the same research history. Autonomous science can thus be studied not only through what machines execute, but through how their actions give rise to reliable, transferable understanding—and how that understanding makes further inquiry possible.

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Methods

Worlds, interfaces and experimental units

ChemWorld compiles compatible process components into persistent virtual chemical environments. The public contract specifies materials, legal operations, instruments, resources and task objectives. Private parameters determine each world's responses. Agents access the laboratory and public records, without evaluator state, simulator code, private parameters or withheld reference outcomes. Operations change samples; measurements may consume material. Candidate transitions are validated before commitment. Rejected operations do not constitute the requested physical intervention, although attempts can consume contract-defined allowances. Exact replay checks environment transitions, not regeneration of model decisions.

The broad study covers electrochemistry, partitioning, reaction and thermal processing, aqueous equilibrium, reaction-coupled crystallization and purification. These are controlled virtual systems, not experimentally validated predictions for named physical substances. The integrated repeated design covers all six systems through complementary matrices. Five fixed worlds recur across arms and repetitions; their parameters, initial states and keyed observation mappings are fixed. Repetitions are fresh model sessions, not independent world draws or controlled model-seed replications. The model interface does not expose a usable sampling seed. Repeated identical interventions can share keyed observations, so they are not independent noise samples.

Integrated matrices and new-session coverage

The integrated coverage design contains 450 planned researcher sessions in four matrices, combining a new 300-session block with 150 retained sessions. The objective–information matrix uses electrochemistry and reaction processing, each with five worlds × three information conditions × two commissions × three sessions, or ninety per system. The process matrix uses partitioning under discovery and purification under optimization, each five worlds × three information conditions × three sessions, plus the retained forty-five equilibrium and forty-five corrected-crystallization sessions. The opportunity matrix comprises thirty new crystallization sessions and thirty retained equilibrium sessions: five worlds × Research/NoResearch × three sessions, all Opaque. Thirty retained matched-world sessions form the fourth matrix. The retained 150 and the historical broad 300 are distinct from the new 300; a shared coverage structure does not make their protocols or periods identical.

New participants use gpt-5.6-sol at medium requested reasoning effort. Source studies allow twelve batches; the fifteen no-research participants receive none. Scientific accounts, twelve-query predictions and retrospectives are sealed in order without reference-truth feedback. The 300 planned sessions yield 288 complete chains, 298 valid forecast stages, 3,379/3,420 source final assays and 898/900 valid sealed stages. Electrochemistry completes 90/90 chains and 1,080 assays; reaction processing completes 88/90 chains and 1,080 assays, with 88 valid forecasts; partitioning completes 45/45 chains and 540 assays; purification completes 35/45 chains and 499/540 assays but retains all forty-five forecasts; crystallization completes 30/30 chains, including all thirty forecasts and 180 research assays. The new source and recommendation records have 285 and 240 verified replay receipts, respectively.

Ten purification sessions retain material-exhaustion shortfalls rather than replacement histories. One reaction forecast is absent after tool-budget exhaustion and another is invalid because its intervals violate the assessment requirements. All corresponding operations and later available accounts are retained. Four explicit service interruptions are retried after the primary queue, under the user's predetermined technical-retry rule; every interrupted attempt remains in the ledger and does not add a scientific unit. Earlier runtime-development attempts are separate from this block. No completed scientific outcome is replaced because of its score. The manuscript compilation performs no model or simulator experiment.

New-matrix measurements and aggregation

Each response retains its native assessment definition. Electrochemistry and purification use a single keyed final observation as reference; partitioning uses noiseless presampling phase fractions; reaction processing uses a five-observation mean for point error and all sixty reference observations across twelve queries for coverage; crystallization uses noiseless preassay responses and initial-charge recovery. Partitioning intervals are nominal 90%, whereas the other systems use 80%. The two partition fractions are complementary, not independent evidence. No cross-response or cross-system scientific-ability score is formed.

Response errors and interval diagnostics are computed per session. Within a condition, available sessions are averaged within each fixed world and the five worlds receive equal weight. The electrochemical commission comparison further averages the three information conditions within each world; all forty-five forecasts per commission are available. The reaction Opaque–Aligned optimization comparison uses fifteen valid forecasts per condition. Other reaction cells with missing forecasts retain fourteen of fifteen and are explicitly marked. Purification uses available forecasts even when the source budget is incomplete. Full-grid estimates remain unavailable when planned responses are missing; available-case values do not impute zero error. Repeated sessions, queries and responses share worlds and are not treated as independent

draws for confidence intervals or significance tests. Three repetitions per fixed cell describe variability, not a population reliability rate.

Operating comparisons use independent retests of sealed recommendations. They are distinct from predictions under new query interventions. Scores and yields retain their public definitions and are averaged separately. Interval width, empirical coverage and, where present in the retained evaluator, the proper interval score are reported alongside point error. For the new crystallization opportunity comparison, the score uses alpha = 0.2 and its legal response scale. Research and NoResearch have the same scientific framing and sealed assessment, with research additionally supplying an experimental budget, experience and recording opportunity. Their difference estimates this complete package, not a specific reasoning operation.

Full reading of the new research histories

All 300 new sessions are read, including the fifteen no-research accounts, all 873 notebook versions and 31,922 public operations. The evidence contains 899 nonempty sealed payloads: one includes an invalid forecast, so there are only 898 valid stages; one forecast payload is absent. Coverage comprises electrochemistry (268 notes; 7,647 operations), reaction processing (293; 7,917), partitioning and purification (262; 13,253), and crystallization (50; 3,105). The retained per-session readings verify 1,363 quotations against public records. Missing notes, unsuccessful tests, rejected requests, terminal revisions and later retrospective admissions remain visible. Reading coverage is not an explanation-success denominator.

For each focal judgment, analysis distinguishes the stated phenomenon, candidate explanation or empirical expectation, its first recorded appearance, prospective consequence, actual committed intervention, measurement, revised standing and later use. A conditional plan is not counted as an executed test. Subsequent predictions are not retroactively inserted into earlier notes. A terminal explanation can support a claim about later prediction, but cannot by itself establish an earlier influence on source actions. Source-time public contracts and the frozen execution semantics distinguish rejected requests, rolled-back transitions and task-quality failures. In particular, a reaction-risk threshold does not imply rollback, and direct heating and controller configuration have different public temperature limits. Quench advances the clock and changes state without additional reaction integration; waiting before quench may continue chemistry. The selected crystallization seed review compares requested and committed amounts, rather than treating repeated rejected requests as dose.

All histories receive semantic reading, while illustrative claims are selected retrospectively for their explanatory relevance. Multiple claims can coexist within a session with different support and outcomes. The analysis does not classify a whole session as a successful abduction, infer private cognition or estimate a common self-confirmation mechanism. Interpretation and source verification are performed within the assistant analysis workflow; there is no independent human coding, blinding or second-rater reliability estimate. The earlier action-linked windows and selected explanation histories below retain their original definitions and denominators.

Historical and retained evidence layers

The historical broad pool contains 240 Sol-medium campaigns and 60 additional equilibrium campaigns from Luna, Terra, GPT-5.5 and Astra, requested at medium reasoning effort. The five-model equilibrium comparison includes fifteen Sol campaigns already in the 240. Requested model labels do not establish equal realized computation. The original 240 include 60 electrochemical, 30 partitioning, 60 reaction, 45 equilibrium, 30 crystallization and 15 purification campaigns. They retain 3,597/3,600 final assays, all 720 assessment stages and 238 protocol-conforming campaigns under their original rules. The additional sixty retain 720 batches and 240 stages. Two original source-completion failures remain included. Earlier selected notebook studies comprise another 24 sessions and are not pooled into the broad or retained equilibrium–crystallization process cohort.

The retained equilibrium–crystallization process cohort combines the 45 equilibrium sessions from the earlier 48-session block and its 42-session extension with a complete replacement block of 45 crystallization sessions under a consistent recovery description. Each system covers five worlds, Opaque/Aligned/MisIndexed information and three sessions per cell, with twelve research batches per session. Equilibrium retains 39 complete chains and 41 valid forecasts; corrected crystallization retains 41 and 42, respectively. The combined retained denominator is 90 sessions, 80 complete chains and 83 valid forecasts. Old crystallization sessions remain protocol-affected historical evidence. They are not added to the replacement sessions when calculating current outcomes or process frequencies.

Corrected crystallization has 510/540 source batches, 125/135 assessment stages, 145 notebook versions and 42 recommendation retests. All 45 source trajectories and 42 retests pass replay. Four incomplete chains remain: three lack forecasts, and one retains a valid forecast but lacks the subsequent retrospective. The equilibrium subset and the separate opportunity block retain their own failure tables. Missing outcomes are not set to zero, and later favourable outcomes do not replace completed results. Evidence integration does not change the original development or qualification status of any block.

Information, autonomy and recording

Historical goal comparisons pair independent discovery and optimization campaigns within world, information arm, batch budget and prior locus. Electrochemistry contributes thirty pairs across twelve- and twenty-four-batch conditions; reaction processing contributes thirty across parameter- and structural-prior blocks at twelve batches. Operating outcomes are independent retests of sealed recommendations. The joint-outcome counts use public-score prediction MAE, whereas the reaction prior comparison uses the mean across its six normalized response errors. These outcomes are not interchanged. Budget comparisons likewise concern independent sessions with complete resource envelopes; they do not estimate the effect of continuing the same investigation.

The historical five-model equilibrium comparison contains five worlds, three information arms and one twelve-batch campaign per cell: seventy-five campaigns, including the fifteen Sol campaigns already counted in the broad study. The 1 mM threshold descriptively summarizes the lowest positively loaded nominal concentration assayed; it is not a discovery criterion. Concentration is based on nominal recipe loading, and the three dilute assessment queries are shared across configurations. Forecast errors, original prediction intervals and acquired evidence are retained for all arms. The selected Astra and electrochemical trajectories are retrospective process illustrations, with scientific accounts written after experimentation; they are not contemporaneous notebook records or additional replications. Complete historical design, response tables and case values remain in Supplementary Information.

Opaque supplies the public task and operational context without the additional instance dossier. Aligned supplies partial information appropriate to the system. In equilibrium, this includes an effective-acidity interval and an anchored local dilution relationship. MisIndexed uses another world's archival relationship with the same format and qualification. In crystallization, Aligned gives nominal descriptors for anonymous materials; MisIndexed swaps solvent S1/S3 descriptors while preserving S0, S2 and catalyst descriptors. Neither dossier specifies the complete world law.

The original equilibrium Opaque/Aligned sessions precede its MisIndexed extension. Corrected crystallization is a later complete block. These descriptive comparisons are not all concurrently randomized treatments, and new versus old crystallization is not an estimate of the causal effect of correcting the prompt. The retained equilibrium opportunity comparison uses a common contemporaneous protocol, described below.

Agents choose operations, quantities, measurements, repetitions and grouping within fixed budgets. Equilibrium permits twelve source batches, 180 operation attempts, twelve intermediate instruments and twelve final assays; crystallization uses its fixed twelve-batch materials envelope and 720-attempt ceiling. A notebook role asks for an initial testable expectation when available; subsequent updates are agent-selected, without a write-every-batch requirement or automatic reinjection. The role may influence research and is not assumed passive. Notes retain version and public-event cursor. These public records do not expose private cognition. Absence of a notebook-read tool call does not establish absence of access to the conversation's earlier content.

Assessments and recovery definition

After source research and any operating recommendation, agents seal a scientific account, answer twelve fixed new-intervention queries without laboratory tools and then reflect without truth feedback. Equilibrium includes a structured supplemental assessment. The evidence notation K1/Q/K2/EQS denotes these stages. Queries are hidden during research; post-assessment proposals are not treated as executed source experiments.

Equilibrium errors compare predictions with the mean of five reference observations per query. Coverage checks all five observations. Crystallization uses the noiseless full-process response immediately before final-assay sampling; source observations are noisy. The corrected block retains sixty reference batches, numerically matching the earlier reference outcomes, and separately retests each available committed recommendation.

The corrected recovery definition is seed-excluded recovered solid target divided by the tracked initial reagent charge, in moles, clipped to [0,1]. Source instructions, prediction prompts, observations and scores now refer to this same full-process quantity. The internal crystallization-stage metric instead divides by target product before separation. Earlier process instructions described this latter denominator while observations and scoring used initial reagent charge. Original scores are preserved as protocol-affected diagnostics; they do not enter the corrected cohort. That corrected block was rerun in full rather than retrospectively relabelling those scores.

Quality requires particles, purity at least 0.80 and fines at most 0.50; joint recommendation success also requires corrected recovery at least 0.10. Corrected joint counts are 13/15 Opaque, 10/14 Aligned and 10/13 MisIndexed among available retests, against fifteen planned sessions in each arm. Particle size is the public index min(d50/250 micrometres, 1); its error is not measured in micrometres.

Research paths and process interpretation

We examine abductive inquiry through publicly stated candidate explanations and relevant subsequent tests, distinguishing these from deductive expectations, empirical support and later application. The focal equilibrium interpretation connects an initial expectation, discrepant observations, a proposed alternative, its anticipated consequences and an executed test.

We separately inspect whether the implemented intervention can discriminate the proposed alternatives. Scientific accounts and accurate forecasts alone do not establish this sequence. These records support a case-based process interpretation, not a validated scale of abductive ability or an estimate of mechanism prevalence. The 92 action-linked windows describe recording coverage in the retained equilibrium–crystallization process cohort; they do not count successful abductions or contain the focal opportunity-study case.

For the retrospective explanation-history panel, selection is fixed to the focal equilibrium opportunity session and the four corrected-crystallization cases already illustrated in the preceding manuscript: lower charge, interpolation, and the two contrasting seed accounts. We trace seven related judgments across these five sessions, retaining all notebook versions, including terminal revisions. Fields distinguish the phenomenon, a candidate or expectation actually stated by the agent, first appearance, conditional prediction, executed operations, observations, subsequent standing and later use. Missing conditional statements are left missing rather than supplied by the analyst. The equilibrium loading expectation, component hypothesis and background interpretation are connected strands of one inquiry, not three independent discoveries. The panel is not sampled to estimate frequencies; no explanation-level prevalence or reliability denominator is inferred from the window counts.

The focal operation-semantics check uses the retained execution version and trajectory's mechanism identity, not current runtime assumptions. That world's single non-catalytic feed is selected both by the reagent operation and by component 0; both add to the same initial-amount ledger read by the equilibrium response calculation. Switching these commands at equal loading therefore does not switch chemical identity. Final nominal concentrations in Extended Data Figure 5 divide committed added moles by total added solvent volume; the staged-dilution batch is plotted at its final 0.8 mol/L. No new simulator calls are required for this inspection. Proposed low-loading and orthogonal measurements remain unexecuted in the retained source history. The scope of the resulting plateau is assessed independently of the agent's proposed mechanism.

Retrospective path extraction covers sixty historical electrochemical, forty-five equilibrium and forty-five corrected crystallization sessions. Fifty within-world condition contrasts retain goal, budget, information and cohort. Counts use committed operations; solvent coverage includes additions during resuspension. Rejected attempts do not establish successful material use. Interrupted sessions remain in the planned denominator.

Structured extraction considers all 178 adjacent pairs among 268 notebook versions in the retained 90 sessions. Eligibility requires an earlier note, at least one new committed final assay before the next note, and a committed research operation after that later note; measurement and termination alone do not satisfy continuation. All 178 pairs contain new assays. Ninety-two have the required continuation: 34 equilibrium and 58 crystallization windows in 30 and 35 sessions, respectively. The other 86 pairs (44 equilibrium, 42 crystallization) have no later committed research action and remain available in the evidence archive. The 25 sessions without a complete window are not classified as having learned nothing.

Eligibility is computed from event timing before semantic plan matching, without testing whether a proposed action succeeds. Nevertheless, the rule conditions the sample on continued committed research and excludes terminal revisions from this particular action-linked denominator. All 92 eligible windows are coded. Labels concern one selected immediate plan, not every proposal in a note: 83 matched, seven partly matched and two without an explicit next plan; corrected crystallization contributes 52 matched and six partly matched. Full notes retain abandoned, rejected and altered proposals even when another selected action matches. All 182 update/plan quotations and their assay/action cursors are verified against source records. The archive includes every adjacent pair, selected code, full note and execution history.

The same analyst performs coding and internal scientific review. There is no independent second-rater agreement estimate, blinding or claim of adjudicated mechanism prevalence. The team has assigned review to this workflow; completed internal review is distinguished from independent replication. Counterexamples include warranted retractions, correct recognition of unexecuted tests and accurate but explicitly conjectural extrapolation. Selection of illustrative cases is retrospective.

Matched chemical worlds and explanatory histories

The additional prospective block contains five parameter pairs, two chemical networks and three fresh Sol-medium sessions per world: thirty planned sessions. Network A couples weak-acid dissociation to direct free-ion precipitation; B adds neutral aqueous ion pairing. Within each pair, acid dissociation, solubility-product, cation, activity and initial-state parameters are matched, along with the public operational contract. Two actual starter assays, each using 6×10^{-7} mol in 0.080 or 0.075 L, are supplied before autonomous research. The paired noise-free responses differ by no more than one declared final-assay standard deviation in each channel. Participants receive chemical framing and loading, dilution and scale dimensions, but not the private network identity or parameters. The initial notebook therefore follows supplied observations, not an evidence-free pretest.

Qualification of the implemented candidate covers 230 physical states and assays, thirty exact replays and 120 runtime mean checks. Five bounded nuisance refits do not adequately fit network B with the tested direct-network family; this check does not prove identifiability against all alternative explanations. An earlier failed physical candidate remains separate. These checks concern the world's implemented information structure, not agent learning.

Each session has twelve autonomous batches followed by a sealed scientific account, twelve query forecasts and a retrospective without truth feedback. The queries are withheld during research, although agents can visit corresponding physical

conditions. All thirty sessions complete all stages: 360 source final assays, ninety sealed stages and 1,080 query-response predictions. Eighty notebook versions comprise twenty sessions with three notes and ten with two. Thus continuous mid-point revision cannot be assumed in every session. Three fresh sessions per world share fixed observation keys; the model sampling seed is unavailable.

Every committed source action sequence is replayed in its own and the paired world: thirty source replays and sixty own/partner trajectories pass. Separation is the absolute difference between noise-free response means divided by the final-assay standard deviation: 0.002 for normalized pH and 0.006 for each fraction. Of 360 final assays, 326 separate at least one channel by three standard deviations and 158 separate at least two; all thirty sessions include an assay of the latter kind. This threshold describes physical information, not statistical significance, intended discrimination or successful abduction. Counterfactual replay fixes actions and does not estimate how an agent would adapt to different observations.

Forecast errors use the mean of five retained reference observations per query; intervals are evaluated against all five observations, which are shared across sessions in the same world. Session values are averaged within each world, then the ten worlds equally. Intervals are nominal 80% prediction intervals, not confidence intervals for means. Their high empirical coverage must be read alongside their widths and does not by itself establish calibration. This block has no no-research arm and uses different chemical laws from the opportunity comparison.

All eighty notes and ninety sealed stages are read with their executed actions. Sixty-seven selected focal histories retain 365 verified quotations. A separate retrospective reading covers 105 existing sessions and 309 notes, tracing 112 selected histories and 218 note anchors; four selected sealed follow-ups add thirteen anchors. The 105 are the retained ninety plus fifteen research-opportunity sessions, not additional experiments. Histories can overlap within a session and include terminal revisions. They are descriptive units selected for explanatory relevance, not independent discoveries or a population rate of abduction. Initial expectations, candidate explanations, empirical relations, executed tests and subsequent uses remain distinguishable; absent causal or conditional statements are not supplied by the analyst.

Source operations retain 1,901 attempts: 1,893 committed, six rolled back and two rejected by validation. Another 131 recovered tool failures are counted separately. The first launch attempt contains four startup failures, twenty-six unstarted sessions and no scientific operations; after a thread configuration repair, the entire thirty-session block restarted. All earlier failures and resource records are retained. No completed scientific outcome is replaced because of its value.

Source-only forecast references

Two transparent references are computed for each valid corrected-crystallization forecast using only that researcher's successful source final assays. The first averages all available assays. The second averages assays whose exact set of committed solvent identifiers matches the query, falling back to the overall mean when none match. Solvent sets include successful solvent additions and resuspension, excluding rejected actions. The fitting function receives no test truth. These references are descriptive calculations, not separately trained agents or mechanistically complete baselines.

There are 42 valid sessions and 504 queries. Same-solvent fallback occurs in 186/504 queries: 1/180 Opaque, 95/168 Aligned and 90/156 MisIndexed. Matching does not control thermal history, quantities or final liquid composition. References have no predictive intervals. Each response is reported in its own public scale; no combined score is formed across recovery, purity, size and fines.

The added distribution analysis uses those same 42 valid sessions and their 504 source final assays. For each response we expose source observations, all query reference means, forecasts, signed errors and session summaries. Descriptive pooled means and quantiles retain repeated query targets; they are not independent-condition estimates. Signed error is forecast minus reference mean. The world-equal signed mean first averages twelve queries within each session, then sessions within each world and finally the five worlds. Three missing forecasts remain in the planned denominator of 45. These calculations use only the corrected cohort and do not import purity levels or directional bias from historical crystallization.

Retained equilibrium research-opportunity comparison

The fixed design contains five equilibrium worlds, Research/NoResearch and three fresh Sol-medium sessions per cell, for thirty planned sessions. All use Opaque information. The shared public background contains the task and measurement definitions, material catalogue, legal operations/instruments and initial state; the instance archival prior is null. Opaque therefore means absence of an additional instance-specific law, not absence of public chemical framing or model pretraining. The complete initial public contexts accompany the manuscript. Fifteen within-world/repeat checks confirm scientific-input equality after removing only the non-scientific campaign identifier. Research additionally receives twelve autonomous source batches and the existing notebook opportunity; NoResearch receives no source operations or observations. Both arms use identical assessment prompts, schemas, order and restrictions.

All thirty sessions reach terminal status. Research completes 180/180 batches; the combined arms complete 117/120 assessment stages and provide 29 valid forecasts. In world three's third Research repetition, the forecast stream disconnects after the scientific account. Q and the two later stages are missing, without replacement. Thus Research has fourteen valid forecasts, NoResearch fifteen; Research world three has two rather than three. Fifteen source trajectories and twenty-five

reference trajectories (300 reference batches) pass exact replay. The retained resource ledger is complete for 29/30 sessions; one has only recorded partial usage. Recorded input plus output totals 13,428,656 tokens. Missing usage is not zero and monetary cost is unavailable.

A first execution attempt failed because the assessment transport configuration omitted a required field. Four launched sessions, two completed source histories totalling twenty-four batches and their resource costs are retained separately; no scientific-account stage completed. After fixing that execution defect, the whole fixed thirty-session block restarted. This was a protocol repair, not replacement based on scientific outcomes. Only the complete restarted block enters the comparison.

The comparison estimates the total research-opportunity package under a shared assessment. It does not isolate autonomy relative to supplied data, feedback alone, notebook use or private representational change. It is between fresh sessions, without a within-session pretest. World and arm order are balanced in a predetermined launch schedule, not a random sample of world laws.

Quantitative summaries and missingness

Equilibrium macro MAE equally weights normalized pH, dissociation fraction and precipitation signal across the designated queries. The extreme-dilution group comprises three queries fixed before the repeated blocks; the remaining nine are not assumed entirely in distribution. Session errors are averaged within each world, then the five worlds receive equal weight. Individual sessions and all missing stages remain visible. Responses, queries, reference observations and batches are correlated within sessions and worlds; they are not independent researcher samples.

Coverage and mean interval width accompany the proper interval score for the agents' nominal 80% intervals[22]. With lower endpoint l, upper endpoint u, outcome y and alpha = 0.2, the score is (u-l) plus (2/alpha)(l-y) when y is below l, or (2/alpha)(y-u) when y is above u. Lower is better. Aggregates weight worlds equally. Agent intervals are not confidence intervals for our group means.

A post hoc sensitivity analysis assigns the single missing opportunity forecast any macro MAE in [0,1], the legal error range, and recomputes means using the planned three sessions per arm/world and five equal worlds. Research-minus-NoResearch bounds are [-0.098659, -0.031992] overall and [+0.036768, +0.103434] for dilution. These are conservative bounds on missing-outcome influence, not imputation, confidence intervals or population inference. Three repetitions per fixed condition describe variation without establishing a population reliability rate.

Matched quench intervention

A separate retained block applies the eighteen original first-cohort crystallization recommendations in five worlds, with and without a single quench operation: ninety pairs, 180 trajectories. All other action arguments and initial states are matched. Quench is removed at its recorded location or inserted immediately before the first seed/crystallization-cooling operation. No agent adapts or receives these outcomes.

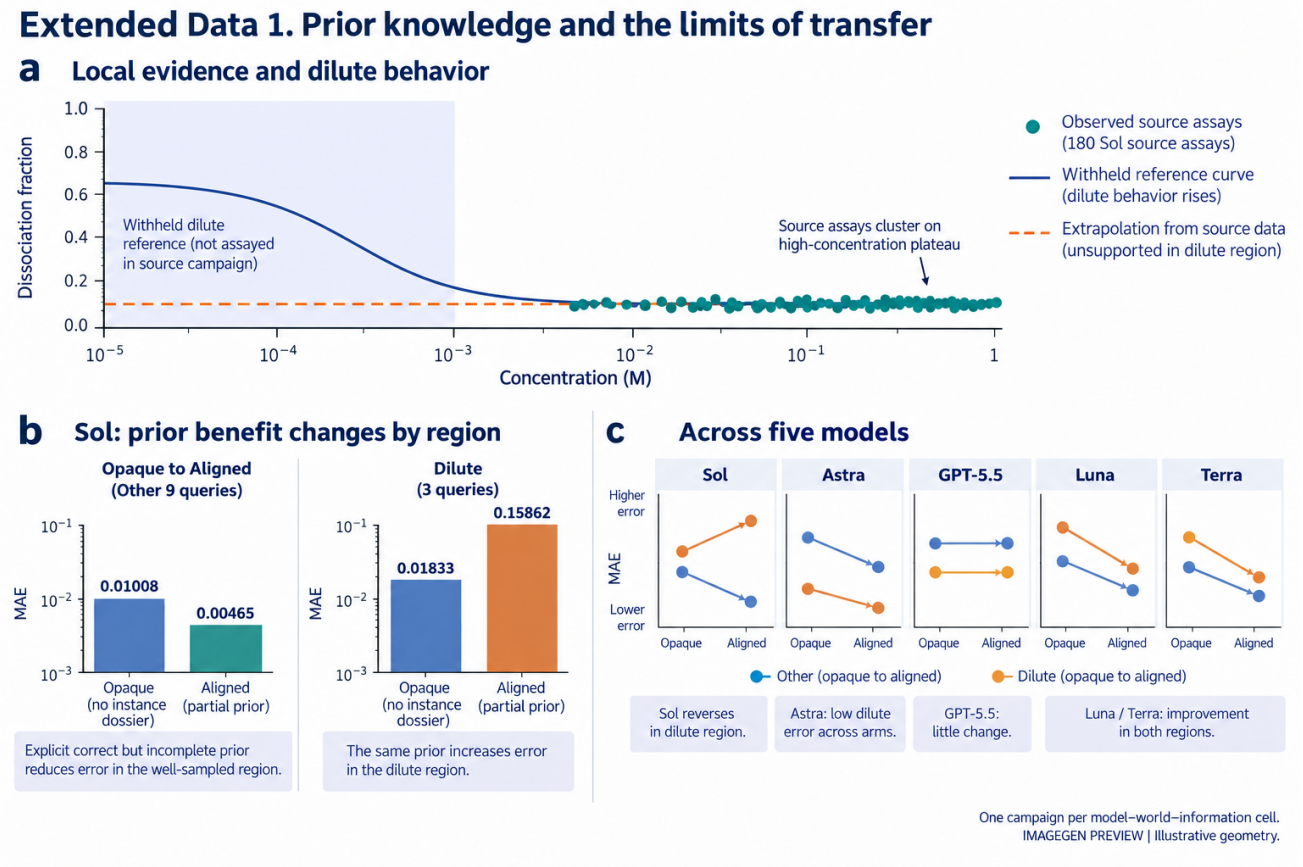
All trajectories are retained and replayed; seventy-five pairs reach both final assays. In five, the absent arm lacks particles; in ten, the present arm makes the subsequent fixed cooling request invalid. Failed arms are not repaired. Among complete pairs, mean present-minus-absent fines changes by +0.30937 and recovery by +0.00741; joint success changes from 34 to 4. These conditional effects describe physical operations under the implemented initial-charge recovery metric. Quench changes multiple physical state variables and does not isolate cooling rate or a cause of scientific judgment. Donor recipes are historical and are not sampled from the corrected block.

Data and code availability

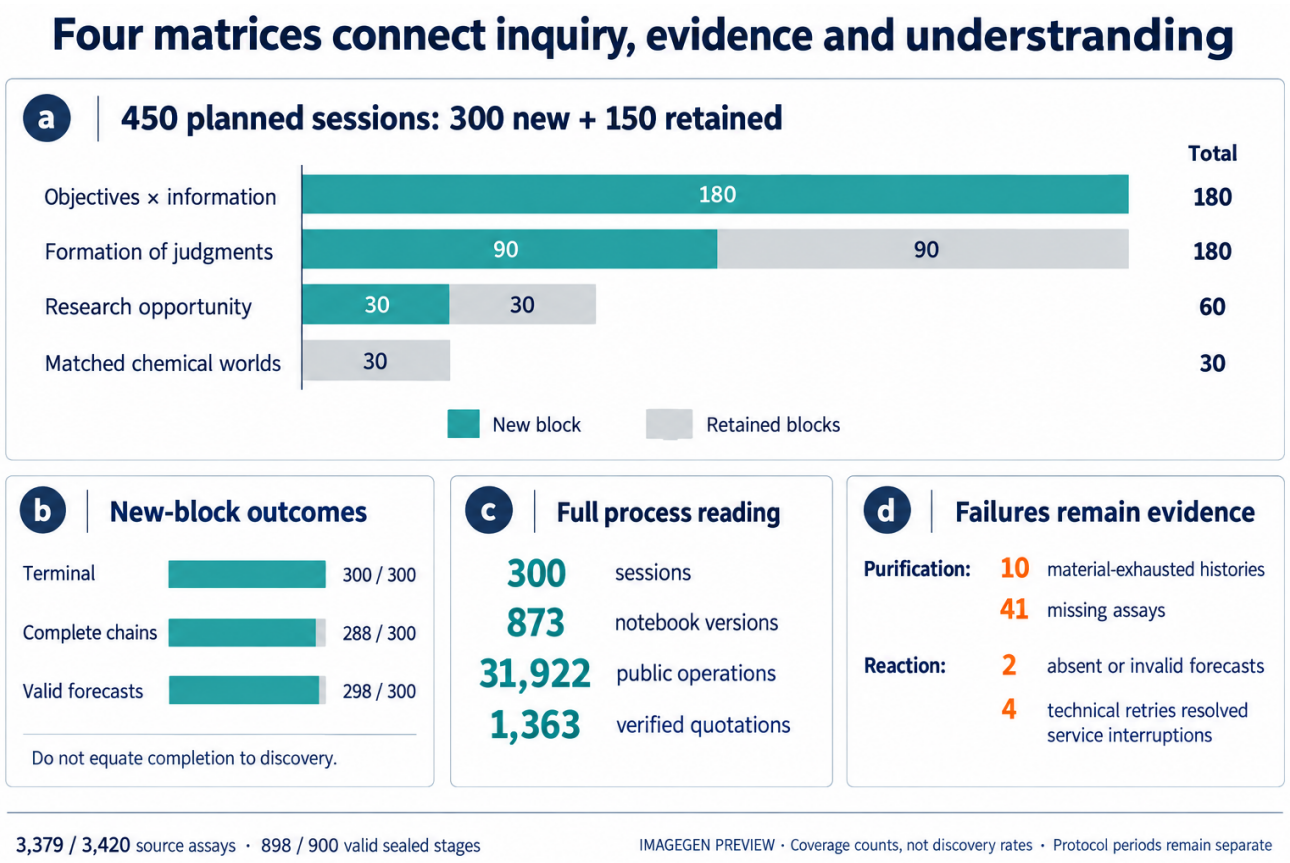
A portable review archive contains the new 300-session machine summaries, all 1,380 planned session–response rows, recommendation outcomes, failures, full curated public histories, semantic readings and quotation anchors. It includes all five system-level coverage tables and the four-matrix design. The retained equilibrium opportunity data, all 178 adjacent note pairs from the older ninety-session process cohort, corrected-crystallization distributions and thirty matched-world histories remain separately accessible. Missing responses remain explicit. Public scientific payloads are included; raw provider envelopes and credentials are excluded. Original replay and resource records remain in the project evidence locations. A persistent public deposit remains to be finalized.

All six main figures and six Extended Data figures are ImageGen previews. Numerical labels refer to retained evidence, but point positions, curves and distribution shapes are illustrative, not exact observations or statistical uncertainty. New quantitative tables are authoritative for the new matrices; older native artwork is retained for its own historical cohorts. The preview images must be replaced by data-rendered figures before submission. Supplementary Information distinguishes every block and preserves all thirty-four historical numerical tables. Compilation reads retained data without changing the experiments or their evidence status.

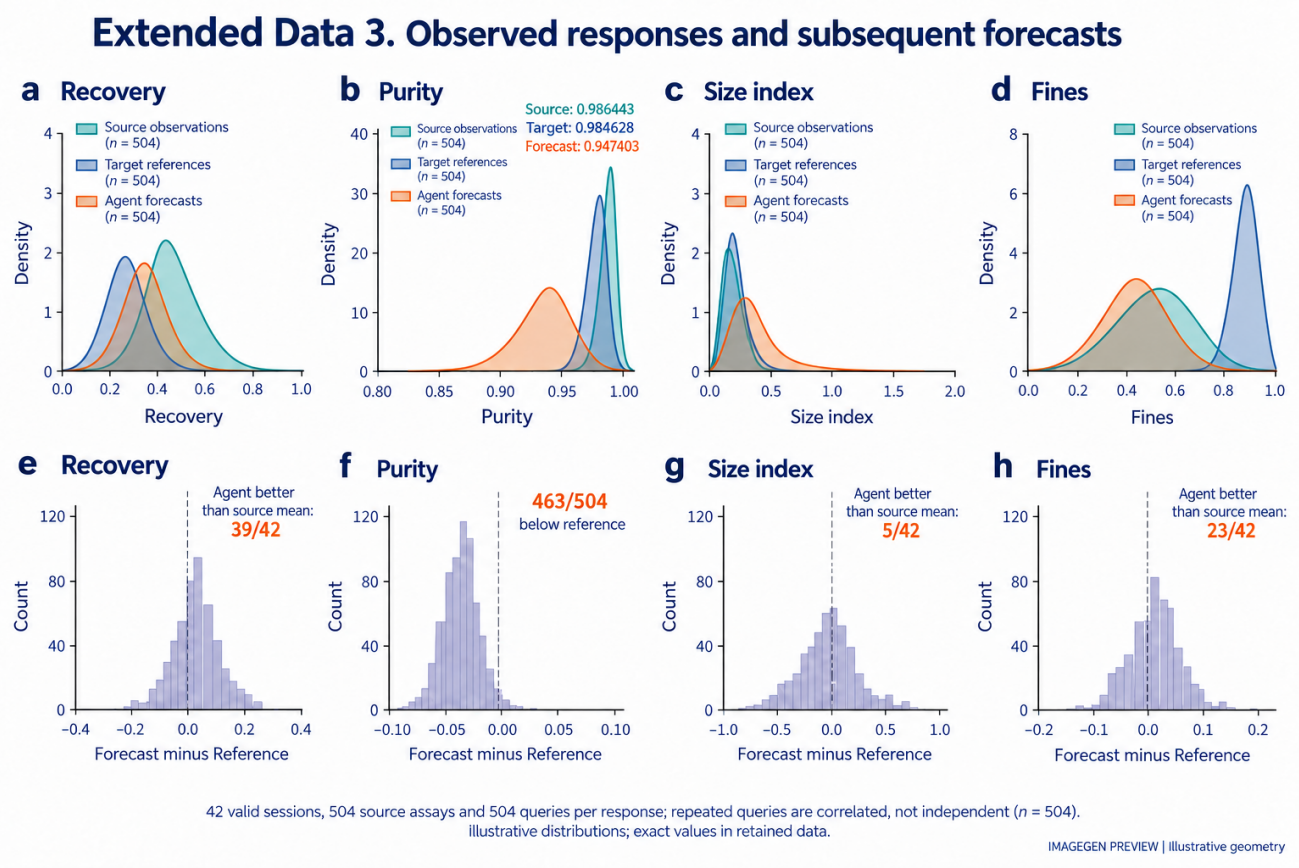
Extended Data



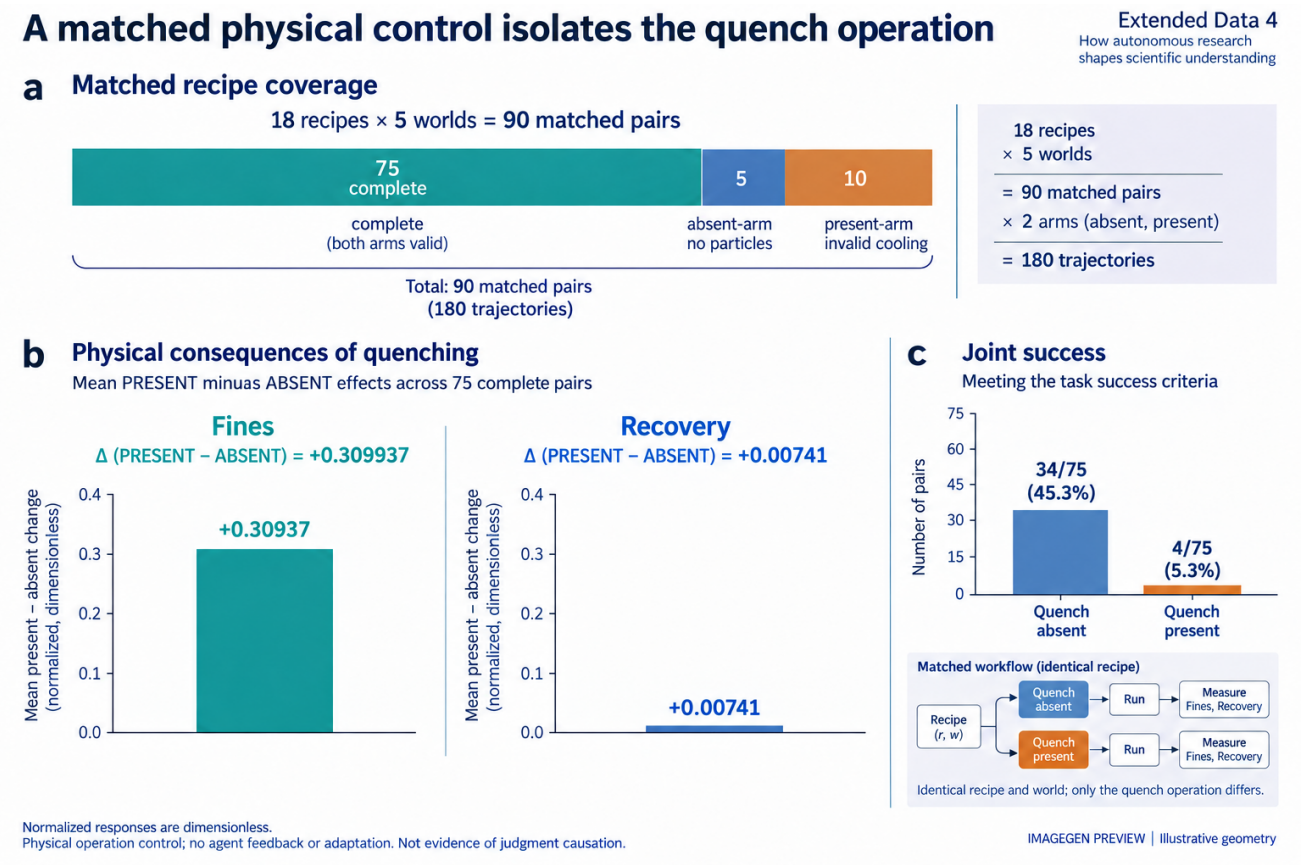
Extended Data Figure 1 | Partial prior knowledge and transfer. a, Schematic local plateau, dilute reference behavior and an unsupported continuation. b, Historical Sol/Opaque versus Aligned errors in two query regions. c, Qualitative patterns across five models, not numerical effect estimates. Historical cells contain one campaign each; the dilute partition was post hoc in that cohort and fixed before subsequent repeated blocks. Repeated-cohort estimates are not pooled here. ImageGen preview: geometry is illustrative; exact values are in the retained evidence.



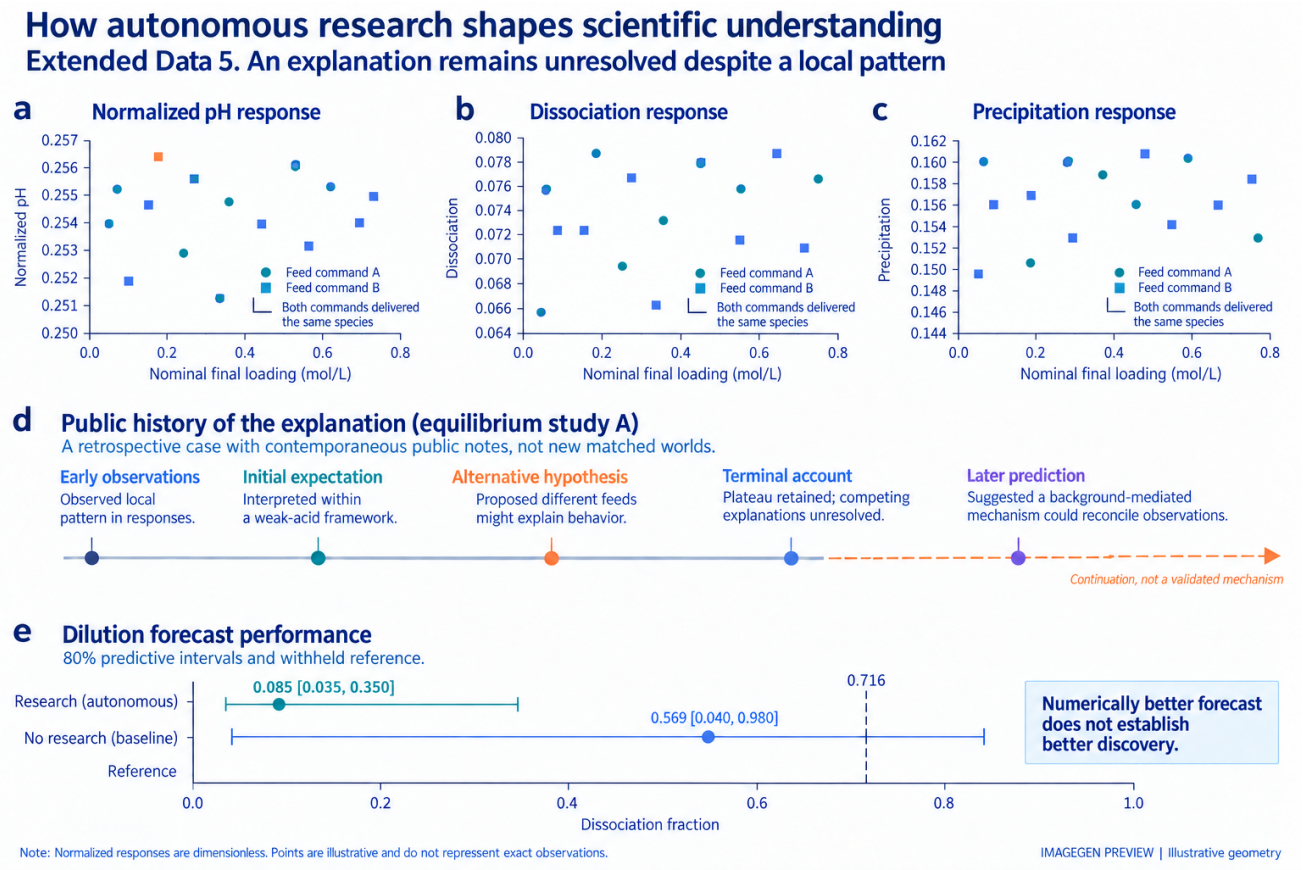
Extended Data Figure 2 | Integrated design and new-block coverage. a, Four matrices contain 450 planned sessions: 300 new and 150 retained under separate protocols. b, The new block retains 288 complete chains and 298 valid forecasts; completion is not scientific success. c, Full reading covers 873 notebook versions and 31,922 public operations, with 1,363 verified quotations. d, Ten purification histories miss forty-one assays after material exhaustion; two reaction forecasts are absent or invalid. Technical retries resolve four service interruptions without replacing scientific outcomes. ImageGen preview: geometry is illustrative.



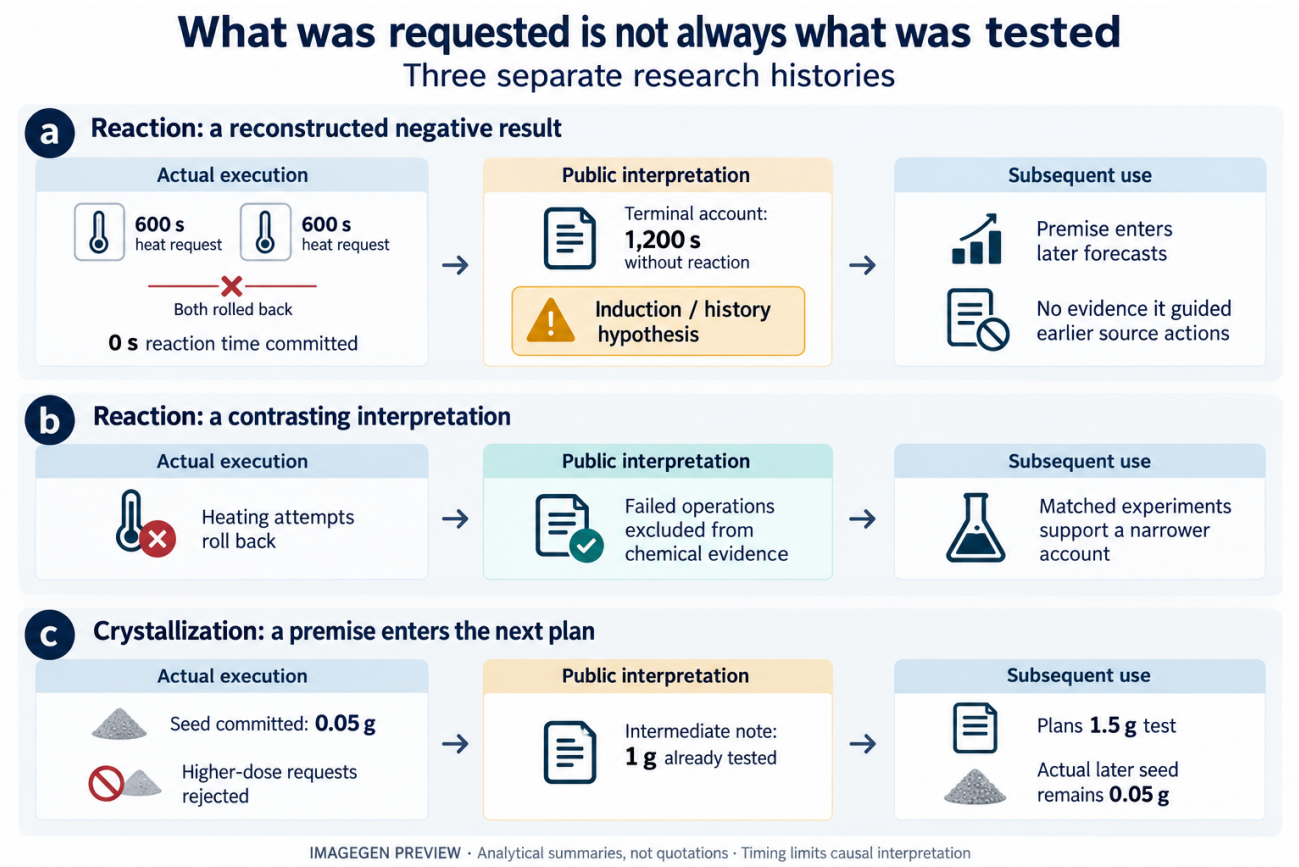
Extended Data Figure 3 | Corrected-crystallization response distributions. a–d, Illustrative distributions for source observations, reference targets and agent forecasts. e–h, Illustrative signed-error distributions; their geometry and bin counts are not observations. The retained cohort has 42 valid forecasts, 504 source assays and 504 queries per response; shared targets are not independent researcher samples. Numeric callouts retain source summaries. Size is a dimensionless index. Three missing forecasts remain in the planned denominator of 45. ImageGen preview: geometry is illustrative; exact values are in the retained evidence.



Extended Data Figure 4 | Matched physical effects of quenching. a, Eighteen historical recipes across five worlds yield ninety pairs, with seventy-five complete, five absent arms lacking particles and ten present arms invalidating cooling. b, Mean present-minus-absent changes among complete pairs. c, Joint task success in those 75 pairs. These are physical effects under the implemented initial-charge recovery metric. Agents neither adapt nor receive these results, so no effect on judgment formation is inferred. ImageGen preview: geometry is illustrative; exact values are in the retained evidence.



Extended Data Figure 5 | A local pattern and an unresolved explanation. a–c, Illustrative source-response patterns in equilibrium study A. Reagent and component commands add the same species in this execution version. d, The public history is condensed from an initial expectation through terminal uncertainty and later use; text is analytical summary, not quotations. e, Retained dilute dissociation forecasts, original 80% predictive intervals and withheld mean. Numerical accuracy does not establish a discovery mechanism. ImageGen preview: geometry is illustrative; exact values are in the retained evidence.



Extended Data Figure 6 | Reconstructing what the experiment tested. a, Two rolled-back heating requests become a claimed 1,200 s negative result in one terminal reaction account; subsequent predictions invoke an induction or history effect. b, Another session excludes rolled-back heating from chemical interpretation. c, An intermediate crystallization note reports a 1 g seed test despite 0.05 g committed, then proposes a higher dose. Timing distinguishes later forecast use from an observed influence on subsequent source actions. Separate selected histories, not randomized correction arms or a common-mechanism estimate. ImageGen preview: text is analytical summary, geometry illustrative.